# Is the Race Decided on Saturday?

An exploratory data analysis of Formula 1, 1950 to 2024

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# 1. Introduction

## Dataset

I used the **Formula 1 World Championship (1950-2020)** dataset by Rohan Rao on Kaggle. It is built from the Ergast API and, despite the name, it is updated past 2020. The version I downloaded covers every race up to the end of the 2024 season. It has 14 CSV files. I used five of them, plus the circuits file for names in the dashboard:

| File | Rows | What it holds |
|---|---|---|
| results.csv (main table) | 26,759 | One row per driver per race: grid, position, points, laps, status |
| races.csv | 1,125 | Year, round, circuit, race name and date |
| drivers.csv | 861 | Driver names and nationality |
| constructors.csv | 212 | Team names and nationality |
| status.csv | 139 | Text label for each finishing status |
| circuits.csv | 77 | Circuit name and country |

I picked this dataset because I am an F1 fan and already had questions about the sport. It is also a real relational dataset, so the tables have to be joined by ID and there is real cleaning work to do.

## Problem definition

The dataset covers every Formula 1 World Championship race from 1950 to 2024: who started where, who finished where, the points scored, and why a driver retired if they did not finish.

**Main question: is the race decided on Saturday?** In other words, how much does qualifying (grid position) decide the result on Sunday?

- How strongly does grid position predict finishing position, and has that link grown stronger since 2010?
- Does one constructor dominate each era?
- Have cars become more reliable (fewer DNFs, fewer mechanical failures)?

## Insights I expect to find

- Pole position wins a large share of races, and the grid matters more in the modern era.
- Each era has one team that wins far more than the rest.
- DNF rates fall a lot over time, mostly because cars break down less.

For the grid vs result questions I use 2010-2024, because the cars and rules in that period are more comparable. For reliability and dominance I use the full 1950-2024 range. All 75 seasons are complete.

# 2. Data Exploration

I first loaded every file with a plain `read_csv` and looked at `head()`, `shape`, `info()`, `describe()` and `isnull().sum()`, first for results.csv and then for the merged table.

### What the raw data showed

- results.csv has 26,759 rows and 18 columns. After joining the other tables by raceId, driverId, constructorId, statusId and circuitId the merged table still has 26,759 rows, so no row was lost or duplicated by the joins.
- `isnull().sum()` said nothing was missing. That was wrong. Missing values are stored as the text `\N`, so pandas read them as normal strings.
- Because of this, `position`, `milliseconds` and `fastestLapSpeed` were loaded as text (object) instead of numbers.
- `fastestLapTime` is text like "1:27.452".
- Grid runs from 0 to 34. A grid of 0 means a pit-lane start or no grid slot.
- Finishing order (`positionOrder`) goes up to 39 and laps up to 200 (the Indianapolis 500 was part of the championship in the 1950s).
- The status table has 139 labels (137 of them appear in the results), from "Finished" and "+1 Lap" to "Engine", "Magneto" and "Collision".

### Missing values after reading \N correctly

| Column | Missing | Why it is missing |
|---|---|---|
| position | 10,953 | Driver was not classified (DNF, DNQ and so on) |
| milliseconds / time | 19,079 | Race time only exists for classified finishers |
| fastestLapTime / fastestLapSpeed | 18,507 | Fastest lap data only exists from 2004 |

# 3. Data Cleaning

All cleaning is done in the notebook (Step 4) and the result is saved as `data/clean_data.csv` (26,759 rows, 32 columns). The steps:

- **Missing values.** Read every file with `na_values=['\\N']`. The before and after table in the notebook shows the counts above appear only after this step.
- **Text to numbers.** Converted position, milliseconds and fastestLapSpeed with `pd.to_numeric`.
- **Lap time.** Turned fastestLapTime ("1:27.452") into seconds (87.452) in a new column `fastest_lap_sec`.
- **Grid 0.** 1,638 rows have grid 0. I kept them but added `grid_clean` where 0 becomes NaN, and used grid_clean for every grid analysis.
- **Name clashes.** Several tables have a `name` column. They became race_name, team and circuit_name. I also built driver = forename + surname.
- **Year range.** Kept 1950-2024 only, so no partial season is included.
- **Duplicates.** `df.duplicated().sum()` is 0. There are 91 cases of the same driver twice in one race. These are real 1950s shared drives, so I kept them.

## New columns

| Column | Meaning |
|---|---|
| finished | True if status is "Finished" or "+n Lap(s)". These drivers were classified. |
| starter | False for did not qualify, did not prequalify, withdrew, did not start, excluded, not classified. |
| dnf | A starter who did not finish. |
| dnf_type | Mechanical, Crash/collision or Other. Built by grouping the 130+ status labels. |
| decade | 1950, 1960 ... 2020. |
| podium / win | positionOrder of 3 or better / positionOrder equal to 1. |

For dnf_type, crash/collision covers Accident, Collision, Collision damage, Spun off, Puncture, Damage and Debris. Other covers labels like Retired, Disqualified and driver illness. Everything else (engine, gearbox, brakes, hydraulics, magneto ...) counts as mechanical.

# 4. Exploratory Data Analysis

All plots use the same style: dark background, red for the main point and grey for the rest. Each title states the finding.

## Univariate

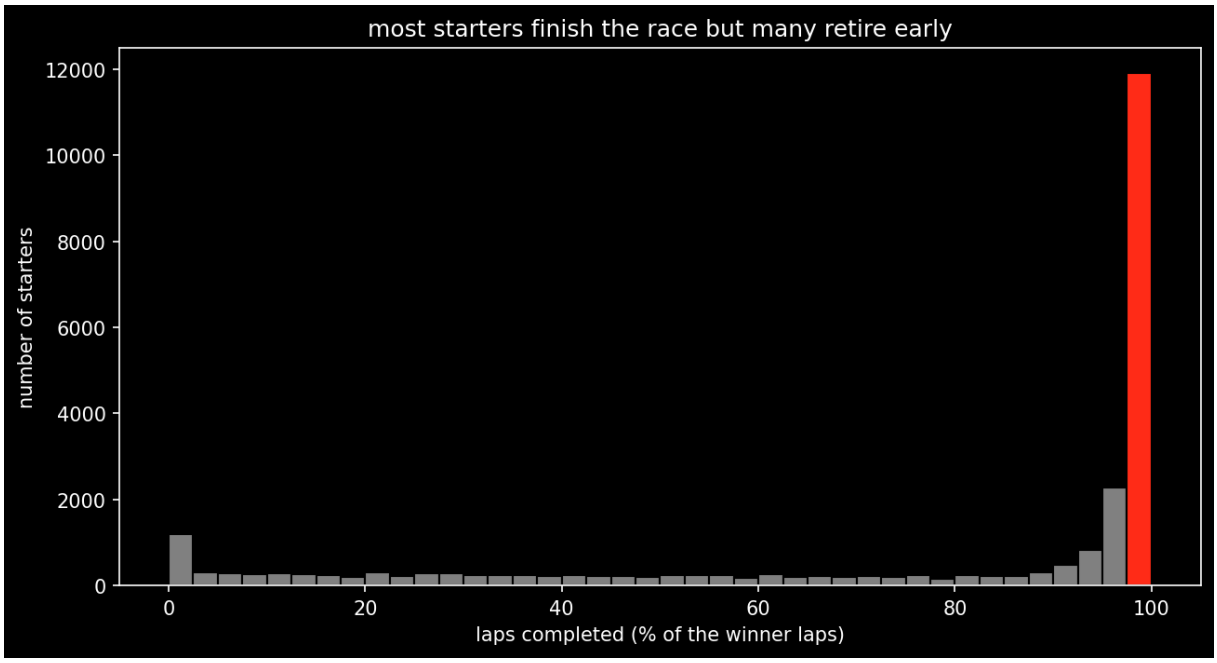



Figure 1. Laps completed as a share of the winner's laps, all starters 1950-2024. About half of the starters reach the end (red bar). The flat tail is retirements, with a spike in the first laps.

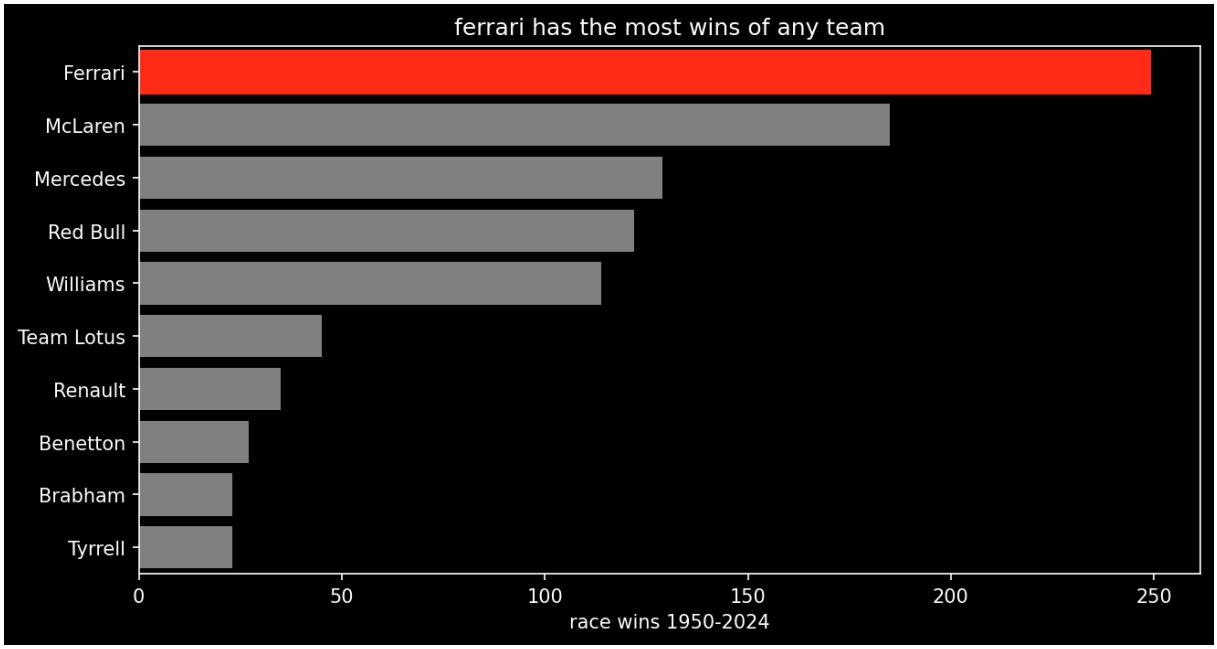



Figure 2. Top 10 constructors by race wins, 1950-2024. Ferrari leads. Mercedes got most of its wins in only 11 seasons.

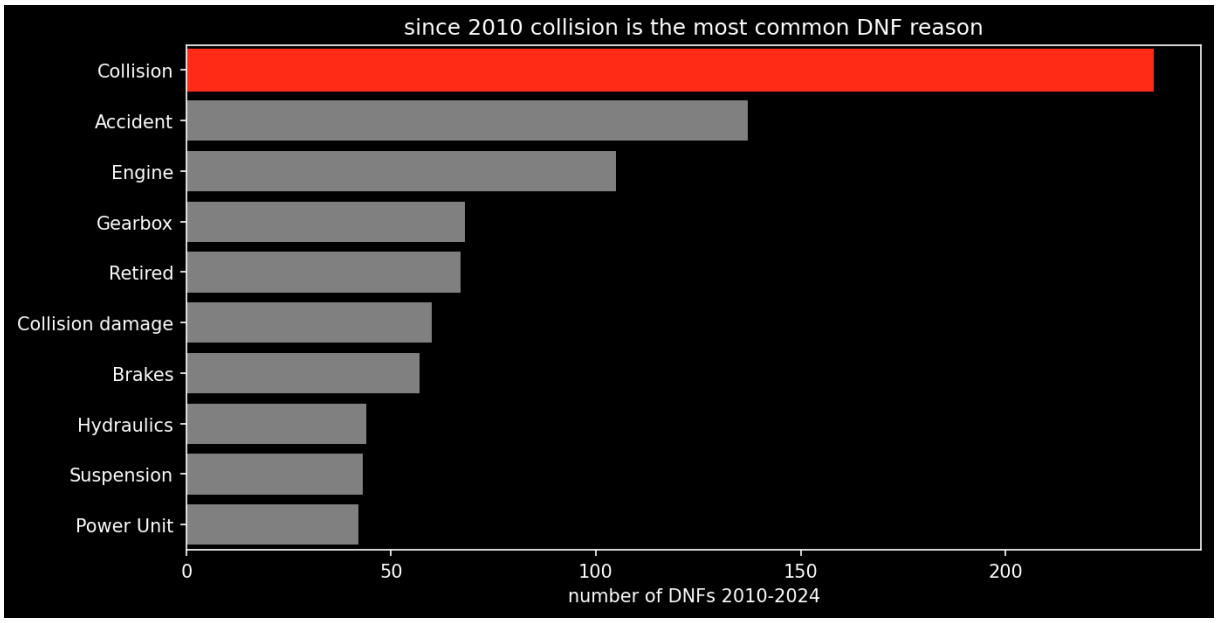



Figure 3. Top 10 DNF reasons, 2010-2024. Collision is the biggest single label. Mechanical problems are split over many labels, so as a group they are still the largest.

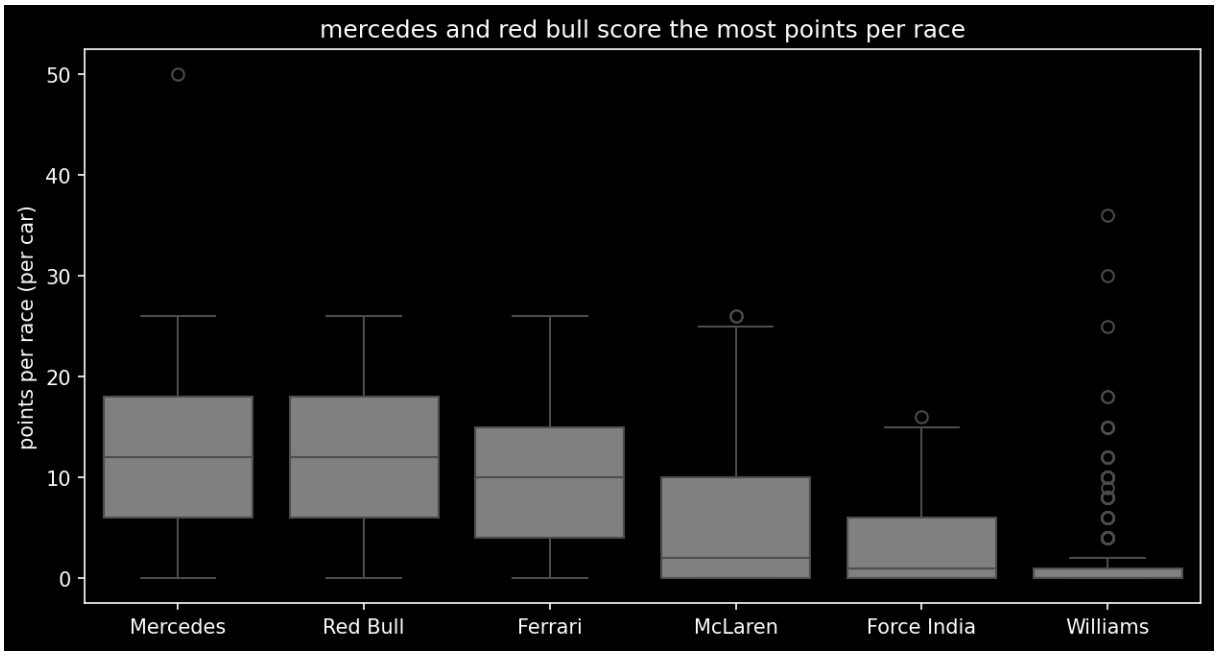



Figure 4. Points per car per race, 2010-2024, top 6 teams by points. Mercedes and Red Bull both have a median of 12 points. The dots at 50 and 36 are the double-points race in Abu Dhabi 2014.

## Bivariate and multivariate

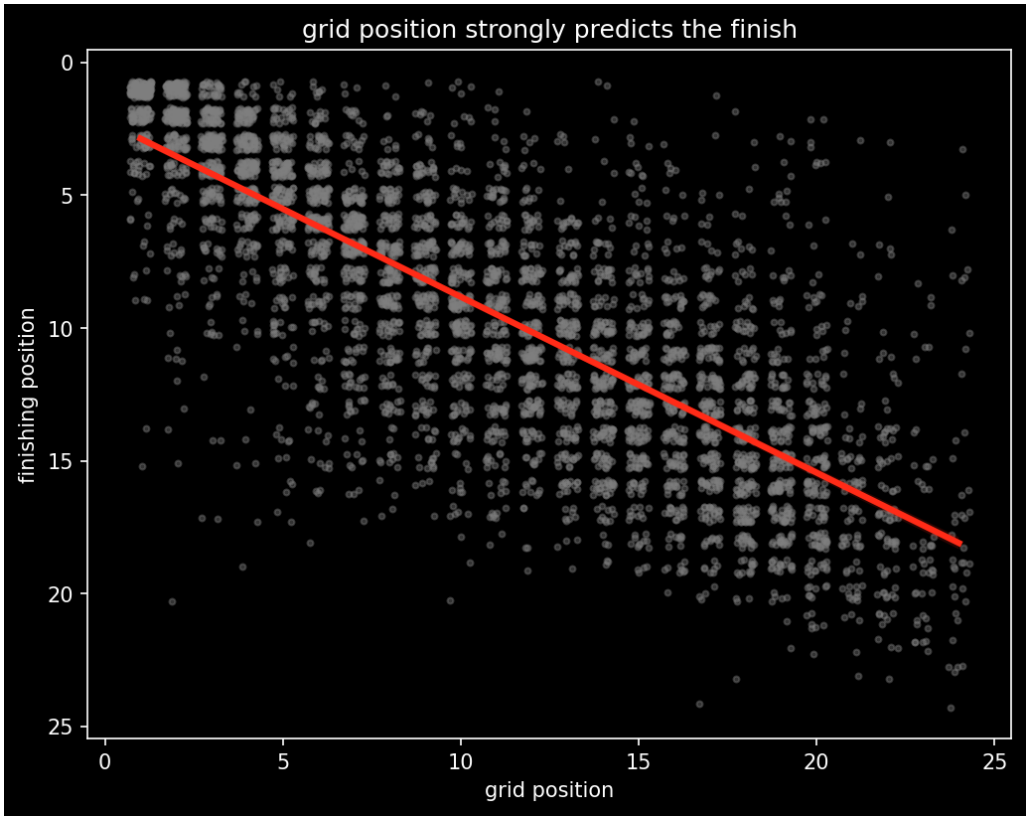



Figure 5. Grid vs finishing position, classified finishers 2010-2024, with jitter and a linear trend. The correlation is 0.78.

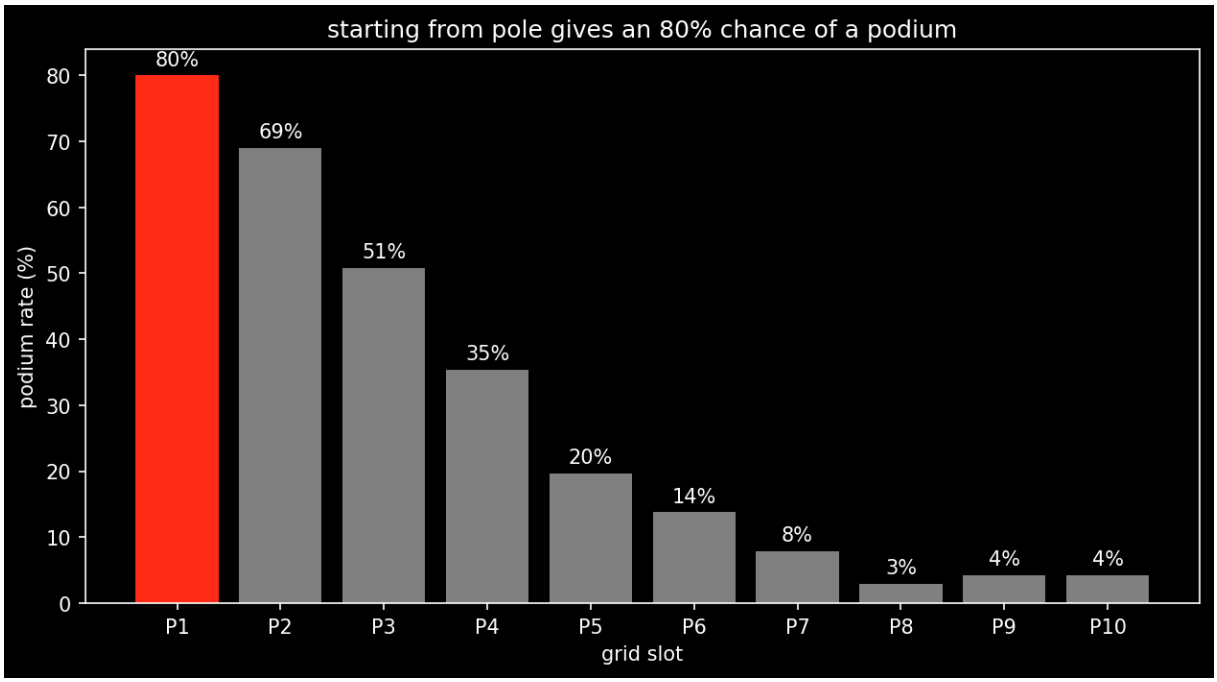



Figure 6. Podium rate by grid slot, 2010-2024. 80% from pole, 20% from P5 and about 4% from P8 to P10.

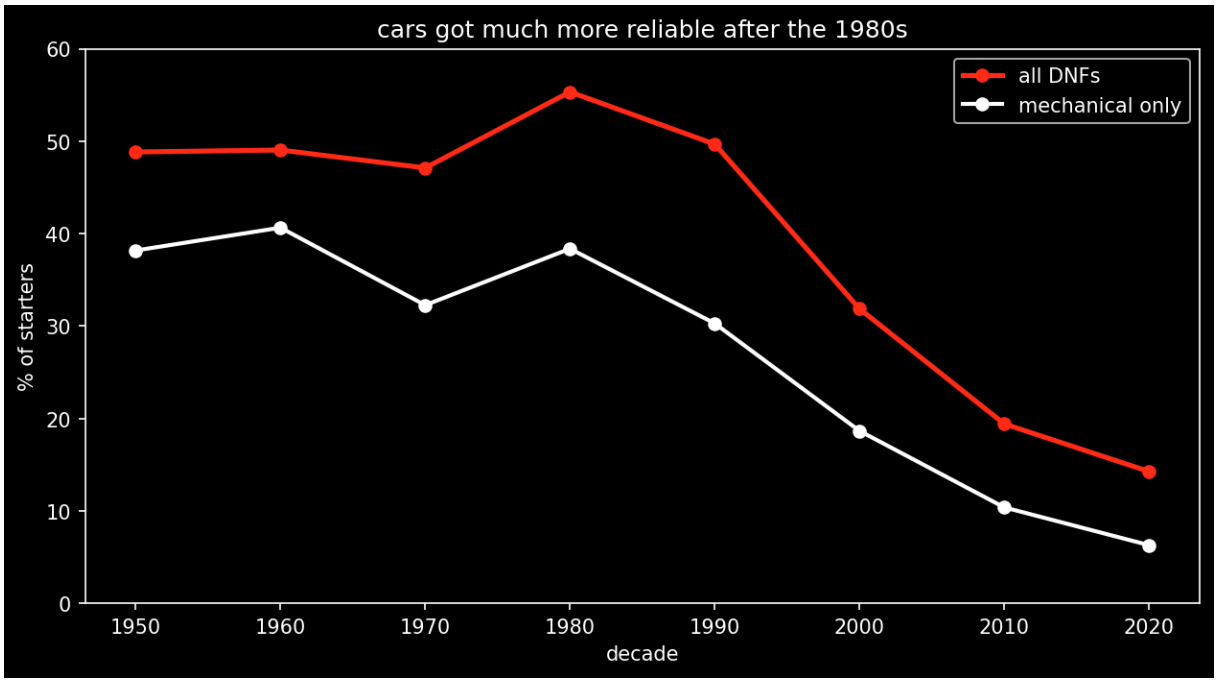



Figure 7. DNF rate among starters per decade (red) and the mechanical part only (grey). About half of starters retired until the 1980s. In the 2020s it is 14%, and mechanical DNFs are 6%.

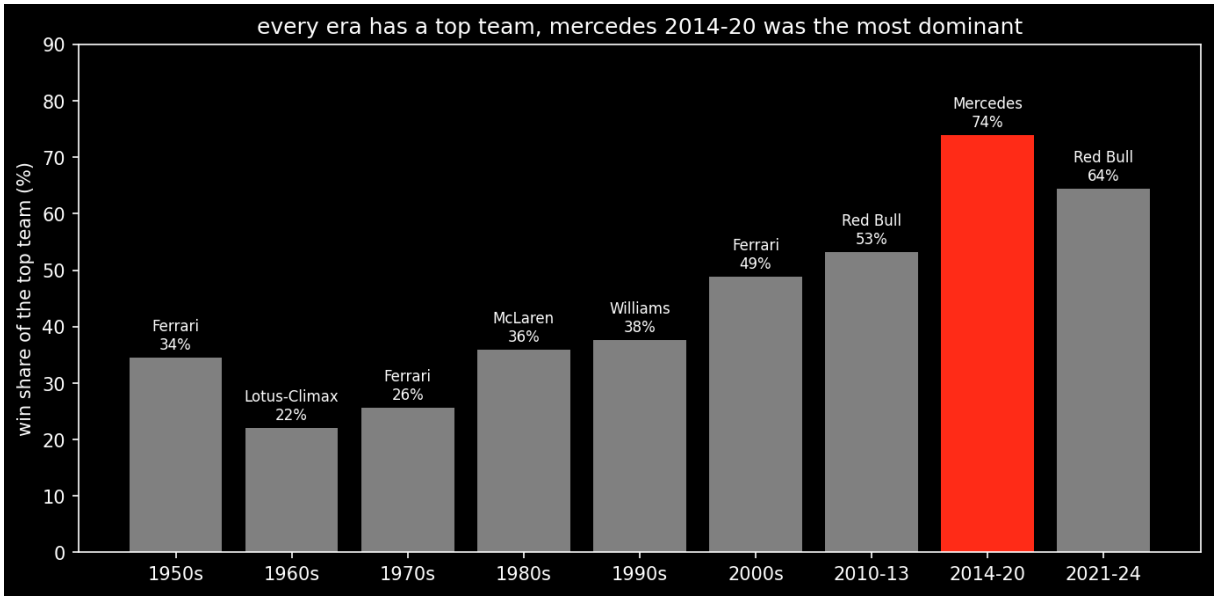



Figure 8. Win share of the top constructor in each era. Mercedes in 2014-2020 (74%) is the most dominant team in the data.

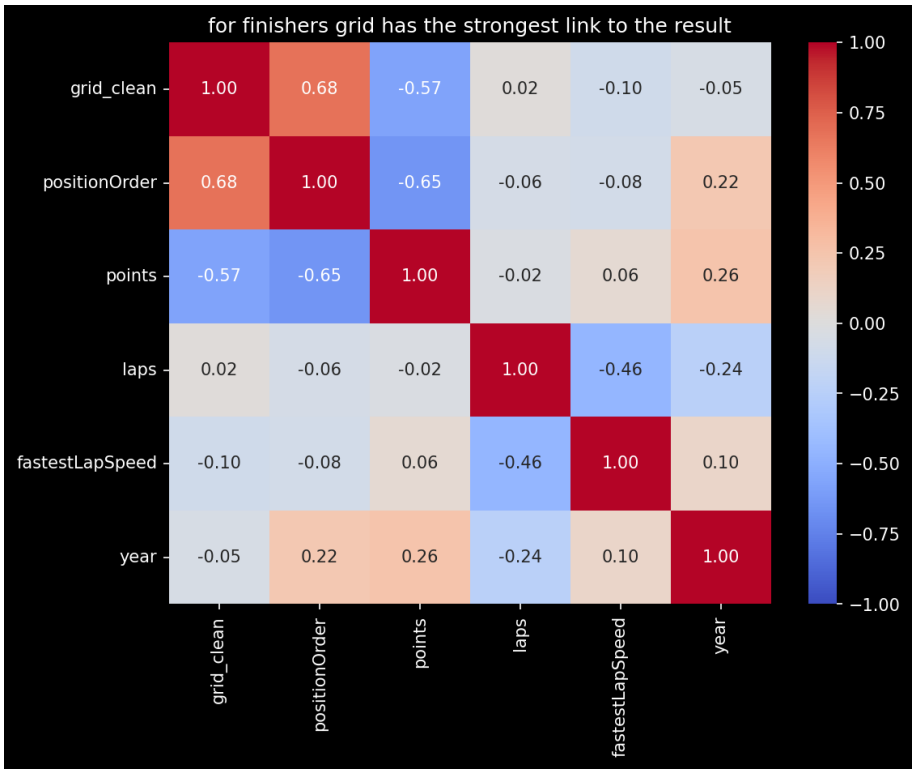



Figure 9. Correlation between numeric columns, classified finishers 1950-2024. Grid vs finishing position (0.68) is the strongest link that is not built into the scoring rules.

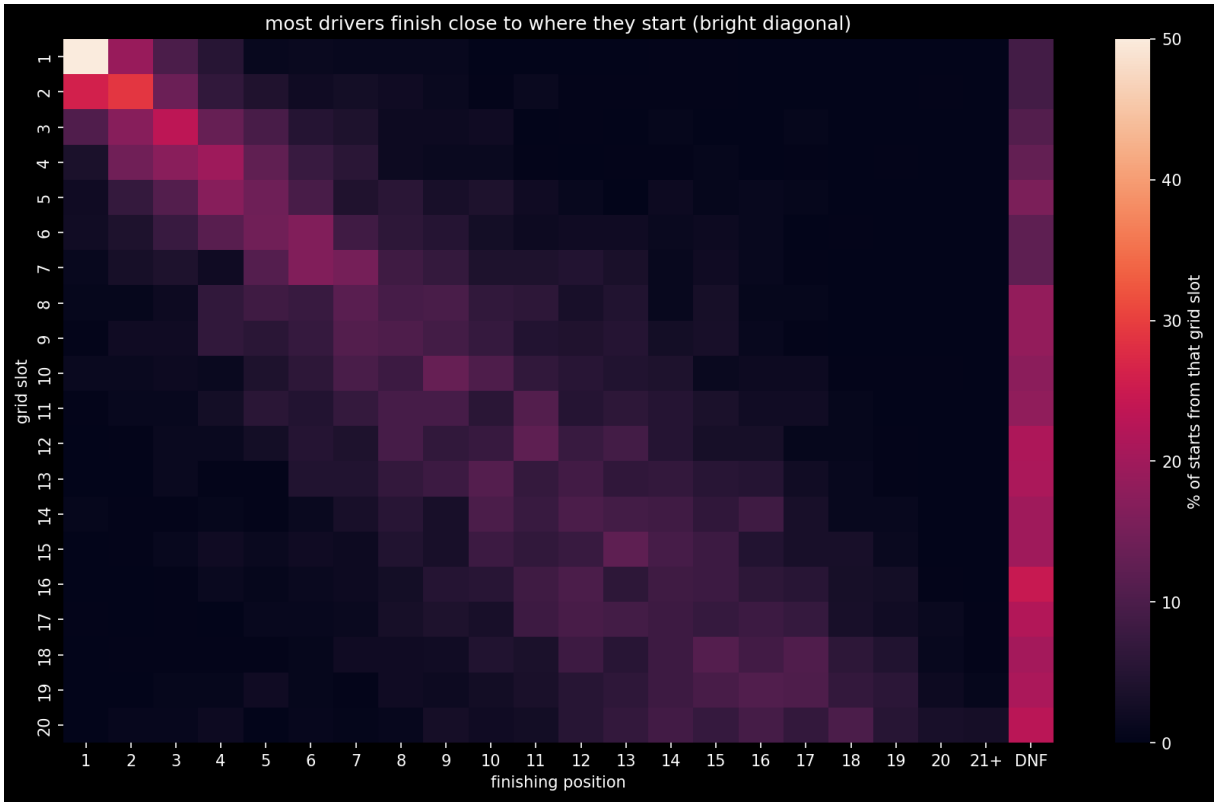



Figure 10. Where each grid slot finished, 2010-2024 (each row adds to 100%). The bright diagonal shows most drivers finish close to where they start. Pole ends in a win 51% of the time.

# 5. Dashboard Overview

The Streamlit dashboard (`app/app.py`) loads clean_data.csv and lets the user explore the same questions. Run it with `streamlit run app/app.py`.

- **Filter bar:** pinned to the top of the page instead of a sidebar, so the charts get the full width. It has a season range slider (default 2010-2024), a constructor multiselect, a circuit dropdown and a reset button. The dataset description is at the bottom of the page.
- **Headline:** the main question with a short answer that changes with the filters, and four stat tiles: wins from pole, wins from the front row, finishers within 3 places of their grid slot, and DNF rate. Each tile shows how far the selection is from the 1950-2024 value.
- **Qualifying tab:** win, podium or points rate by grid slot (switchable), a flow diagram from grid position to the result (win, podium, points, no points, DNF), a grid vs result heatmap and the distribution of places gained or lost.
- **Dominance tab:** a bar chart race of constructor wins that plays season by season, the leading team's share of wins in every season (in team colours), and a table of the top team in each era.
- **Reliability tab:** DNF rate per season (all and mechanical) and what ended the race in each decade.
- **Drivers tab:** pick a driver (or tap a quick pick) to see starts, wins, podiums, pole starts and places gained, rings for win, podium and finish rates, every race they started (grid against result), a radar against the field average, and their career season by season in team colours.
- **Predict tab:** pick a grid slot, team and season and the logistic regression gives a podium chance on an animated gauge, with the chance from every grid slot shown as a bar chart.
- **Play tab:** three F1 mini games built on real grids and results (Lights Out reaction starts, Pit Stop, and Grid Dash, a top-down race from a real grid slot). They can be played solo, by two players on one device, or as an online championship that friends join from their phones.
- **Max Verstappen assistant:** my own chibi drawing of Max Verstappen, animated, floating over the page as a fan-made AI character powered by Claude who talks like Max. He follows the mouse with his eyes, can be dragged anywhere, and a double-click opens a chat window. He answers questions about the data by calling tools that query clean_data.csv, so the numbers come from the data and not from memory.
- **Data tab:** a preview of the filtered rows, summary statistics and a CSV download.
- All charts are animated (drawn with ECharts): they grow in when a tab opens and again when the filters change, and every bar, line and cell shows its numbers on hover. The headline numbers count up. Each tab ends with a short insight written from the filtered data.

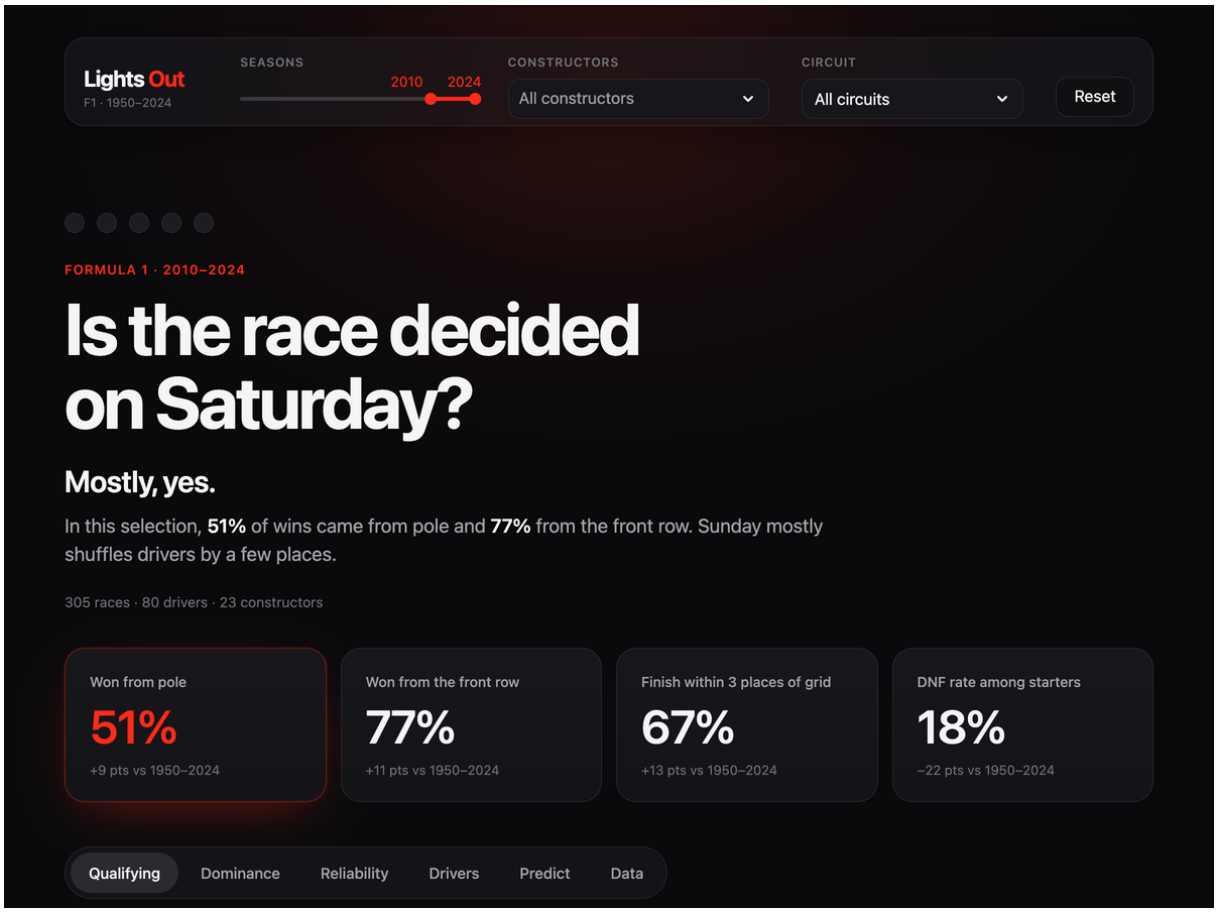



Figure 11. The filter bar, the headline answer and the four stat tiles.

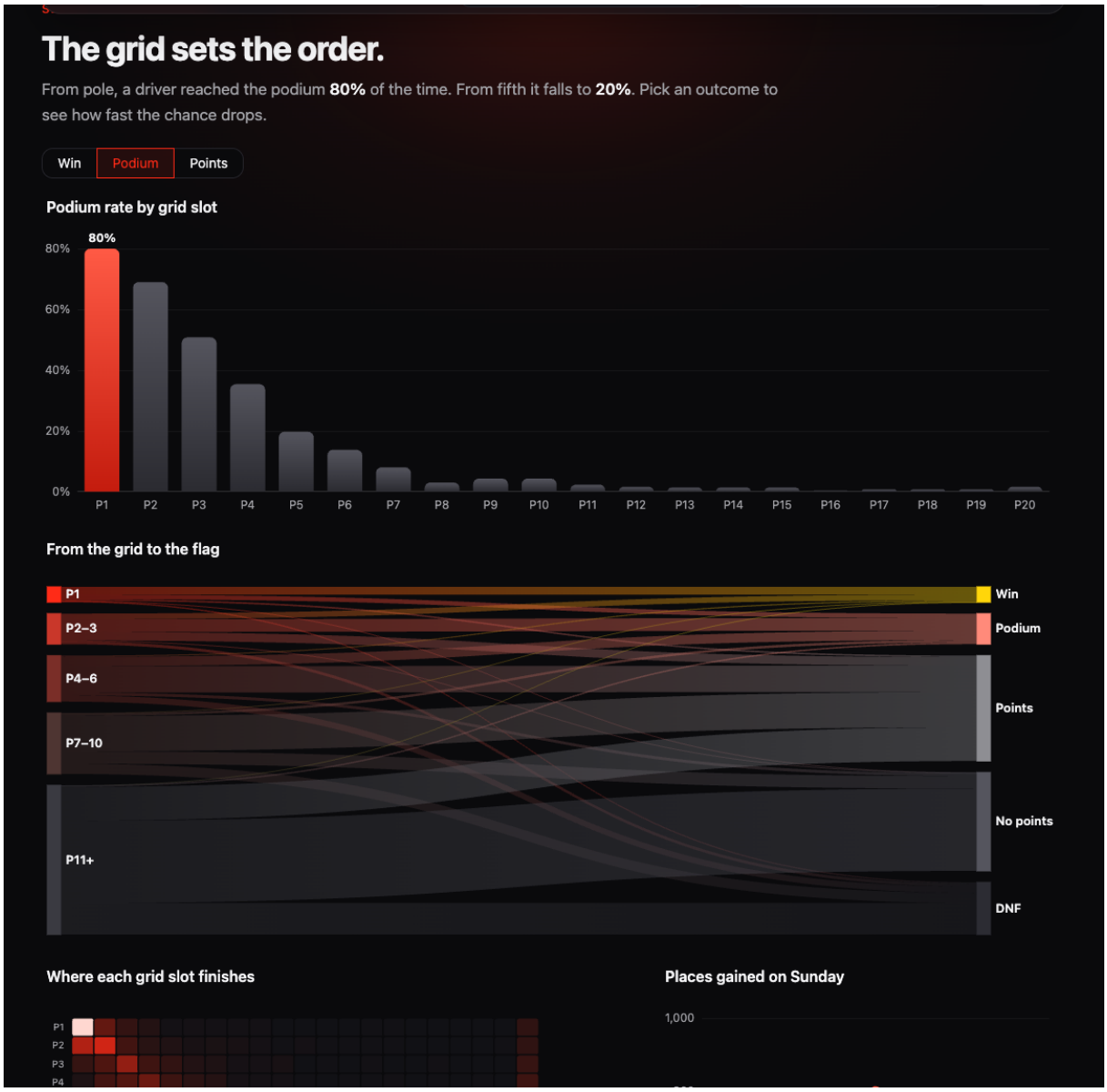



Figure 12. Qualifying tab: rate by grid slot and the flow from grid position to the result.

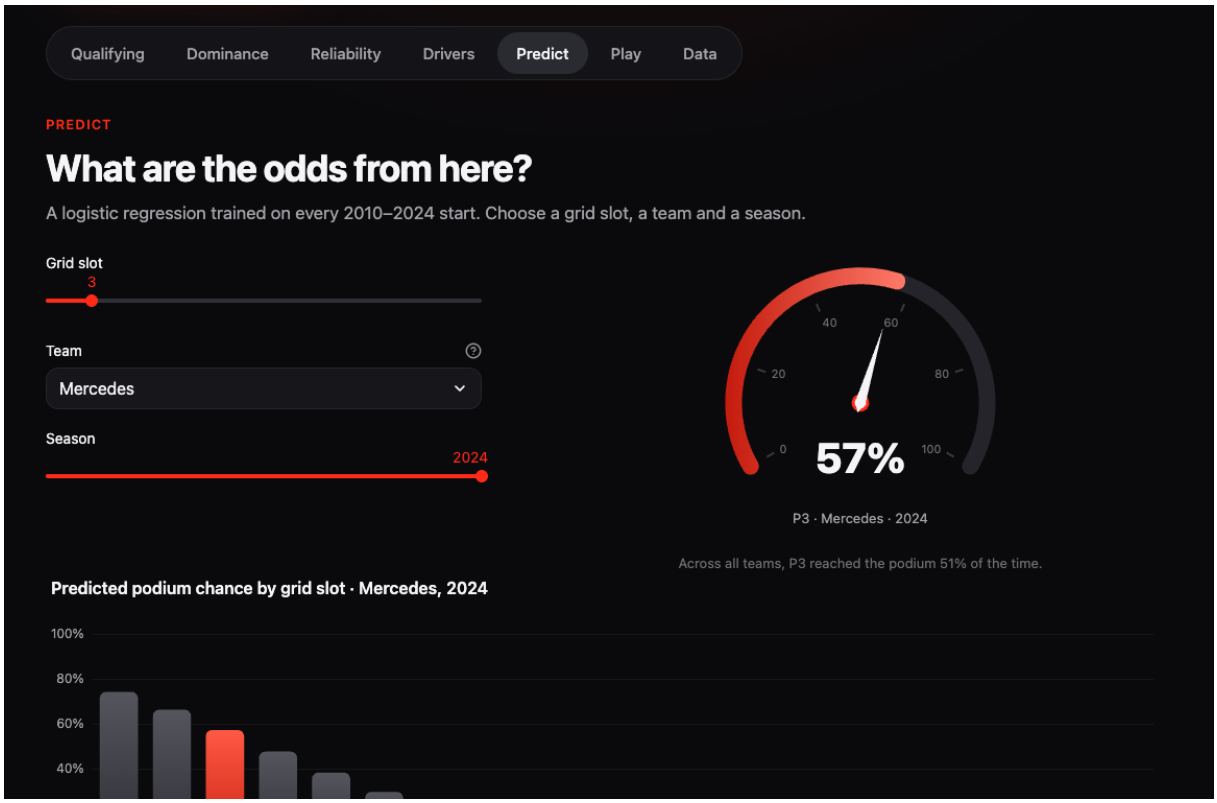



Figure 13. Predict tab: the podium chance gauge for a chosen grid slot, team and season.

# 6. Insights and Conclusion

## Main insights

- **Qualifying decides most of the race.** Since 2010 the pole sitter won 51% of races, the front row won 77%, and only 4 of 305 races were won from outside the top 10.
- **Grid predicts the finish.** The correlation is 0.78 for classified finishers, and 67% of them finish within 3 places of their grid slot.
- **One team rules each era.** Mercedes won 102 of 138 races (74%) from 2014 to 2020. Red Bull won 21 of 22 races in 2023.
- **Cars are far more reliable.** The DNF rate among starters fell from about 49% in the 1950s to 14% in the 2020s. Mechanical DNFs fell from 38% of starters to 6%.
- **Retirements today.** Since 2010, 51% of retirements are mechanical, 41% are crashes or collisions and 8% are other reasons.

## Key numbers

| Stat | Value |
|---|---|
| Rows / races / drivers / constructors / seasons | 26,759 / 1,125 / 861 / 211 / 75 |
| Podium rate by grid P1 to P10, 2010-2024 | 80, 69, 51, 35, 20, 14, 8, 3, 4, 4 % |
| DNF rate among starters, 1950s to 2020s | 49, 49, 47, 55, 50, 32, 19, 14 % |
| Top era winners | 1950s Ferrari 34%, 1960s Lotus-Climax 22%, 1970s Ferrari 26%, 1980s McLaren 36%, 1990s Williams 38%, 2000s Ferrari 49%, 2010-13 Red Bull 53%, 2014-20 Mercedes 74%, 2021-24 Red Bull 64% |
| Most wins, 1950-2024 | Hamilton 105, Schumacher 91, Verstappen 63 |

## Podium model

A logistic regression predicts a podium from grid slot, team (top 6 plus Other) and year, trained on 2010-2024 starters with a stratified 80/20 split. It reaches precision 0.71, recall 0.53, F1 0.61 and ROC-AUC 0.92. A dummy model that always says "no podium" gets 0 for precision, recall and F1, and 0.50 ROC-AUC. Grid slot does most of the work.

## Conclusion

The race is not fully decided on Saturday, but most of it is. Grid position and the car explain most of the result, and Sunday mostly moves drivers by a few places.

## What I learned

- How to join several tables by ID and check that a merge did not lose or duplicate rows.
- Missing values are not always NaN. Here they were the text \N and isnull() missed them.
- How to turn 130+ messy labels into a few groups that answer a question.
- A model needs a baseline. Accuracy alone looks good even for a model that does nothing.

## Limitations

- The points system changed several times, so I compare positions and wins instead of points across eras.
- Status labels are not consistent across decades, so the mechanical vs crash split is approximate.
- There is no weather, tyre or strategy data in these tables.
- Grids before the 1980s had different sizes, so grid slots are not fully comparable across eras.

## Future recommendations

- Add the pit_stops and lap_times tables to see where places are gained or lost during the race.
- Compare street circuits with permanent tracks.
- Improve the podium model with qualifying gaps, driver form and circuit type.

# 7. References


- Rao, R. Formula 1 World Championship (1950-2020). Kaggle. https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020
- Ergast Developer API (original source of the data). https://ergast.com/mrd/
- pandas: https://pandas.pydata.org
- NumPy: https://numpy.org
- Matplotlib: https://matplotlib.org
- seaborn: https://seaborn.pydata.org
- Apache ECharts (dashboard charts): https://echarts.apache.org
- Streamlit: https://streamlit.io
- scikit-learn: https://scikit-learn.org
- ReportLab (used to build this PDF): https://www.reportlab.com
- Anthropic Claude API (Max Verstappen assistant): https://docs.anthropic.com